Mittuniversitetet
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Försättsblad tentamen / Examination cover

2707183

Anonymitetskod / Anonymous code		
S - 0010 - ENN		
Kurskod / Course code: DT171G	Provkod/ Test code: T101	Tentamensdatum / Examination date: 2024-03-12
Kursnamn / Course name: Datateknik GR (B), Inbäddade system för Internet of Things		
Provnamn / Test name: Skriftlig salstentamen		

Skriv din anonymitetskod på varje inlämnat papper
Write your anonymous code on each sheet submitted

Sätt ett kryss (x) för varje inlämnad uppgift
Use an x to indicate which questions has been submitted

Markera nedan med X / Mark below with an X		Poäng / Credit	Lärarens anteckningar / Teacher's notes	Markera nedan med X / Mark below with an X		Poäng / Credit	Lärarens anteckningar / Teacher's notes
1	X	4		16			
2	X	6		17			
3	X	1		18			
4	X	4		19			
5				20			
6	X	1		21			
7	X	2,5		22			
8	X	7		23			
9				24			
10				25			
11				26			
12				27			
13				28			
14				29			
15				30			
Poängsumma / Points	25,5		Betyg / Grade: D	Lärarsign./ Teachers sign			

Fylls i av tentamensvakt / To be filled in by the invigilator

Antal lösa blad/ No. of sheets submitted	Inlämnad tentamen / Submitted exam	Leg kontroll / Control identification	Sign. tentamensvakt / Sign. invigilator
7	0	✓	Lim

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Försättsbladet skall alltid lämnas in även om ingen uppgift behandlats
Examination cover should always be submitted even if no questions are answered

Utskriven 2024-03-08 kl. 09:14:56

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Anonymitetskod/ Anonymus code: S - 0 0 1 0 - E N N

Sidnr / Page no: 01

Lärarens anteckning / Teachers note: 4

(1) $I = 25mA$

$R_1 = 250\Omega$, $R_2 = 150\Omega$ } $R_1 + R_2 = 400\Omega$

$U = R \cdot I$

$U = 400 \cdot 0{,}025$

$U = 10V$

Skriv ej på baksidan! Do not write on the other side!

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Anonymitetskod/ Anonymus code: S - 0 0 1 0 - E N N

Sidnr / Page no: 02

Lärarens anteckning / Teachers note:

2 $R_{cf} = 1 + 1 = 2\Omega$

$R_{cfe} = \frac{cf \cdot e}{cf + e} = \frac{2 \cdot 2}{2+2} = 1\Omega$

$R_{cfeb} = 1 + 1 = 2\Omega$

$R_{cfebd} = \frac{R_{cfeb} \cdot d}{R_{cfeb} + d} = \frac{2 \cdot 2}{2+2} = 1\Omega$

$R_{cfebda} = 1 + 1 = 2\Omega$ final answer.

$2k\Omega$

ok.

b

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Anonymitetskod/ Anonymus code: S - 0 0 1 0 - E N N

Sidnr / Page no: 03

Lärarens anteckning / Teachers note:

$U = 5V$

$R_1 = 100$
$R_2 = 200$ $\Big\}$ $\frac{R_1 \cdot R_2}{R_1 + R_2} = \frac{200 \cdot 100}{200 + 100} = \frac{2000}{300}$

$R = 6{,}67\ \Omega$

$6{,}67^2 / 0{,}25 = 45 / 0{,}25 = 180$ W ?

1

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Anonymitetskod/ Anonymus code: S-0010-ENN

Sidnr / Page no: 04

Lärarens anteckning / Teachers note:

(4) An optocoupler is a component that is used to transfer electrical signals between two isolated electrical circuits by using light.

* electrical isolation: an optocoupler is used to isolate the high-voltage parts of an electrical circuit from the low-voltage parts of the electrical circuit, protecting the sensitive parts of the circuit from the high-voltage disturbances.

* Signal conversion: It is also used to convert electrical signal from analog to a digital and vice versa.

4

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Anonymitetskod/ Anonymus code: S - 0 0 1 0 - E N N

Sidnr / Page no: ~~06~~ 5

6

The difference is voltage source, electrical consumption and speed?

Lärarens anteckning / Teachers note:

/

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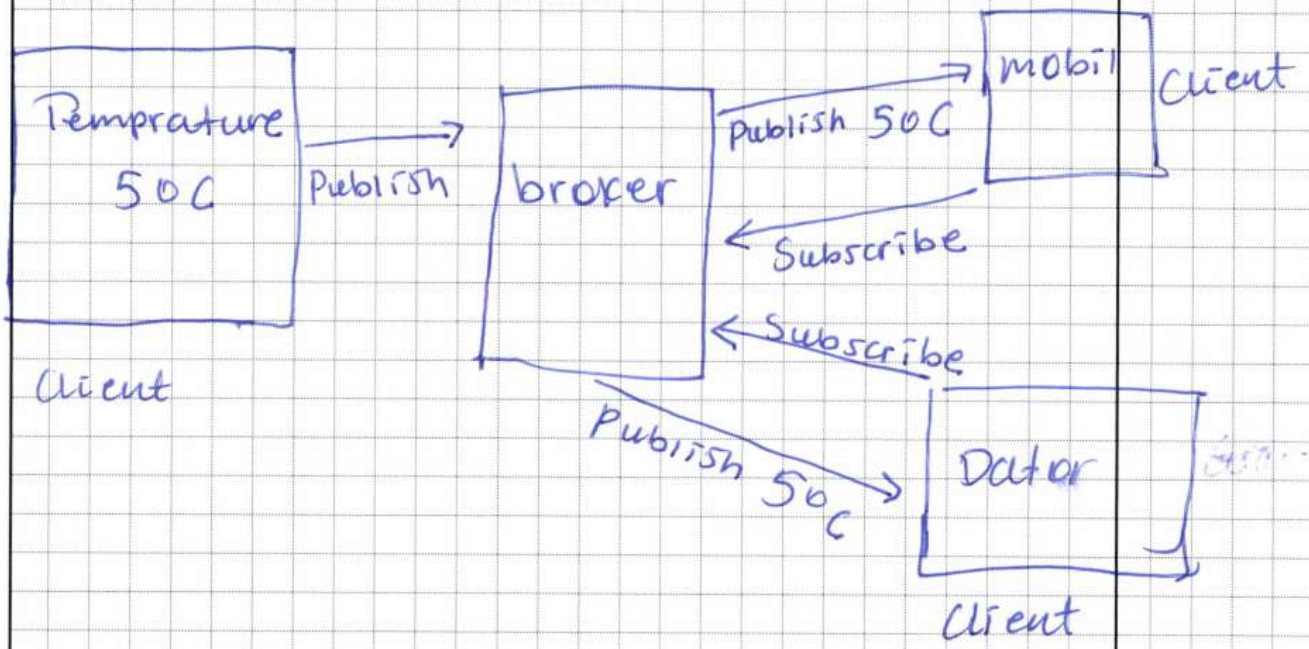
Anonymitetskod/ Anonymus code: S - 0 0 1 0 - E N N

Sidnr / Page no: ~~07~~ 6

Lärarens anteckning / Teachers note:

07

Components are: Clients, Subscribe, publish, topics, security, TCP/IP, port 8080 & broker/server



The MQTT works as the figure shows above, where the clients subscribes or sends subscribtion to the server or broker and the broker publishes the message or the temprature.

2,5

Encryption
Authentication?

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Lärarens anteckning / Teachers note:

8

```
int Sensorpin = A0
Int redLed = 10;
int blueLed = 11;
Int greenLed = 13;
void Setup() {
  Serial.begin(9600);
  Pinmode(redLed, OUTPut);
  Pinmode (blueLed, OUTPut);
  Pinmode (greenLed, OUTPUT);
}
  void loop() {
  int reading = analogRead(SensorPin);
  Float voltage = reading*(5.0/1024.0);
  Float TemprateureC = (voltage - 0,5)*100;
 if (TemprateureC < 15){
      digitalWrite(blueLed, HIGH);
      digitalWrite(greenLed, LOW);
      digitalWrite(redLed, LOW); }
 else if (TemprateureC > 30) {
      digitalWrite(blueLed, LOW);
      digitalWrite(greenLed, LOW);
      digitalWrite(redLed, HIGH); }
 else:  digitalWrite( blueLed, LOW);
      digitalWrite( greenLed, HIGH);
      digitalWrite(redLed, LOW);
```

Component list
Schematics.

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